

Partie 04 — Corrigé

Visualiser des données

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Préparation

```
library(ggplot2)
library(dplyr)
patients <- readr::read_csv("patients.csv")
```

Exercice 1

```
hist(patients$age)
boxplot(patients$tas)
plot(patients$age, patients$tas)
```

Exercice 2

```
ggplot(patients, aes(age)) +
  geom_histogram(binwidth = 5) +
  labs(x = "Âge (ans)", y = "Effectif") +
  theme_minimal()

ggplot(patients, aes(y = tas)) +
  geom_boxplot() +
  labs(y = "TAS") +
  theme_minimal()

ggplot(patients, aes(age, tas)) +
  geom_point() +
  labs(x = "Âge (ans)", y = "TAS") +
  theme_minimal()
```

Exercice 3

```
ggplot(patients, aes(groupe, score)) +  
  geom_boxplot() +  
  theme_minimal()  
  
ggplot(patients, aes(age, tas, color = sexe)) +  
  geom_point() +  
  theme_minimal()  
  
ggplot(patients, aes(age, tas)) +  
  geom_point() +  
  facet_wrap(~ traitement) +  
  theme_minimal()
```

Exercice 4

```
p <- ggplot(patients, aes(age, tas, color = sexe)) +  
  geom_point(alpha = 0.6) +  
  geom_smooth(method = "lm", se = FALSE) +  
  labs(  
    title = "Tension artérielle systolique et âge",  
    x = "Âge (ans)",  
    y = "TAS"  
  ) +  
  theme_minimal()  
  
p  
ggsave("age_tas.png", p, width = 7, height = 5)
```

Exercice 5

1. boxplot ;
2. diagramme en barres ;
3. nuage de points ;
4. nuage de points avec facettes.